# CASE:INC-2026-DFIR-8941-Riyad_Alshehri

- التأكد من قيمة MD5 قبل نقله وبعد النقل الى كالي .

```
PS C:\Evidence> Get-FileHash .\INC-2026-DFIR-8941.raw -Algorithm MD5

Algorithm       Hash
---------       ----
MD5             D2BC2F671BCC9281DE5F73993DE04DF3
```

```
┌──(kali㉿kali)-[~/Downloads]
└─$ md5sum INC-2026-DFIR-8941.raw
d2bc2f671bcc9281de5f73993de04df3  INC-2026-DFIR-8941.raw
```

- البحث عن معلومات الجهاز .

```
└─$ python3 vol.py -f /home/kali/Downloads/INC-2026-DFIR-8941.raw windows.info
Volatility 3 Framework 2.28.3
Progress:  100.00               PDB scanning finished
Variable        Value

Kernel Base     0xf80002605000
DTB     0x187000
Symbols file:///home/kali/volatility3/volatility3/symbols/windows/ntkrnlmp.pdb/3844DBB920
174967BE7AA4A2C20430FA-2.json.xz
Is64Bit True
IsPAE   False
layer_name      0 WindowsIntel32e
memory_layer    1 FileLayer
KdDebuggerDataBlock     0xf800027f60a0
NTBuildLab      7601.17514.amd64fre.win7sp1_rtm.
CSDVersion      1
KdVersionBlock  0xf800027f6068
Major/Minor     15.7601
MachineType     34404
KeNumberProcessors      1
SystemTime      2019-06-29 07:30:00+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductWinNt
NtMajorVersion  6
NtMinorVersion  1
PE MajorOperatingSystemVersion  6
PE MinorOperatingSystemVersion  1
PE Machine      34404
PE TimeDateStamp        Sat Nov 20 09:30:02 2010
```

- تحديد النظام (Windows 7 SP1 - 64 bit) ومعرفة التفاصيل عنه مثل الوقت.

```
em32\winlogon.exe       winlogon.exe    C:\Windows\system32\winlogon.exe
1944    1872    explorer.exe    0xfa80020f7b30  37      854     1       False   2019-06-29 07:28:44.0000
orer.exe        C:\Windows\Explorer.EXE C:\Windows\Explorer.EXE
* 1592  1944    VBoxTray.exe    0xfa80021abab0  13      141     1       False   2019-06-29 07:28:53.0000
em32\VBoxTray.exe       "C:\Windows\System32\VBoxTray.exe"      C:\Windows\System32\VBoxTray.exe
* 2624  1944    DumpIt.exe      0xfa80020a4420  3       45      1       True    2019-06-29 07:29:25.0000
\Desktop\DumpIt\DumpIt.exe      "C:\Users\eminem\Desktop\DumpIt\DumpIt.exe"      C:\Users\eminem\Desktop\
2272    2008    GoogleCrashHan  0xfa80022f0610  7       99      0       True    2019-06-29 07:29:08.0000
s (x86)\Google\Update\1.3.34.11\GoogleCrashHandler.exe  "C:\Program Files (x86)\Google\Update\1.3.34.11\
e\Update\1.3.34.11\GoogleCrashHandler.exe
2284    2008    GoogleCrashHan  0xfa80022f6b30  7       93      0       False   2019-06-29 07:29:08.0000
s (x86)\Google\Update\1.3.34.11\GoogleCrashHandler64.exe        "C:\Program Files (x86)\Google\Update\1.
iles (x86)\Google\Update\1.3.34.11\GoogleCrashHandler64.exe
2700    2692    csrss.exe       0xfa8001cac460  7       164     2       False   2019-06-29 07:29:30.0000
em32\csrss.exe  %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Wi
erverDll=winsrv:UserServerDllInitialization,3 ServerDll=winsrv:ConServerDllInitialization,2 ServerDll=sx
indows\system32\csrss.exe
2728    2692    winlogon.exe    0xfa8002330060  6       121     2       False   2019-06-29 07:29:30.0000
em32\winlogon.exe       winlogon.exe    C:\Windows\system32\winlogon.exe
3012    2992    explorer.exe    0xfa8000eaeb30  28      677     2       False   2019-06-29 07:29:36.0000
orer.exe        C:\Windows\Explorer.EXE C:\Windows\Explorer.EXE
* 2384  3012    VBoxTray.exe    0xfa8000eeeb30  14      144     2       False   2019-06-29 07:29:37.0000
em32\VBoxTray.exe       "C:\Windows\System32\VBoxTray.exe"      C:\Windows\System32\VBoxTray.exe
* 2432  3012    StikyNot.exe    0xfa8000f18b30  10      137     2       False   2019-06-29 07:29:37.0000
em32\StikyNot.exe       "C:\Windows\System32\StikyNot.exe"      C:\Windows\System32\StikyNot.exe
```

```
29 07:28:12.000000 UTC
Sessions        Process: 1696 SearchProtocol started by user 2pac/eminem        N/A     N/A     N/A     2
019-06-29 07:28:12.000000 UTC
Sessions        Process: 2076 dllhost.exe started by user 2pac/eminem   N/A     N/A     N/A     2019-06-
29 07:28:12.000000 UTC
Sessions        Process: 2624 DumpIt.exe started by user 2pac/eminem    N/A     N/A     N/A     2019-06-
29 07:28:12.000000 UTC
Sessions        Process: 2636 conhost.exe started by user /SYSTEM       N/A     N/A     N/A     2019-06-
29 07:28:12.000000 UTC
Sessions        Process: 2700 csrss.exe started by user /SYSTEM N/A     N/A     N/A     2019-06-29 07:28
:12.000000 UTC
Sessions        Process: 2728 winlogon.exe started by user /SYSTEM      N/A     N/A     N/A     2019-06-
29 07:28:12.000000 UTC
Sessions        Process: 2976 taskhost.exe started by user 2pac/SlimShady       N/A     N/A     N/A     2
019-06-29 07:28:12.000000 UTC
Sessions        Process: 3000 dwm.exe started by user 2pac/SlimShady    N/A     N/A     N/A     2019-06-
29 07:28:12.000000 UTC
Sessions        Process: 3012 explorer.exe started by user 2pac/SlimShady       N/A     N/A     N/A     2
019-06-29 07:28:12.000000 UTC
Sessions        Process: 2384 VBoxTray.exe started by user 2pac/SlimShady       N/A     N/A     N/A     2
019-06-29 07:28:12.000000 UTC
Sessions        Process: 2432 StikyNot.exe started by user 2pac/SlimShady       N/A     N/A     N/A     2
019-06-29 07:28:12.000000 UTC
```

- بعد ما عرفت النظام دخلت على فحص العمليات النشطة والجلسات ولاحظت هنا أن فيه أكثر من مستخدم (eminem و SlimShady ) ولقيت برنامج الملاحظات StikyNot.exe شغاله .

```
┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ grep -iE "Users.(SlimShady|eminem).(Desktop|Documents|Downloads)" ~/Downloads/inc_filescan.txt | gre
p -ivE "desktop.ini|\.lnk"
0×3e8ad250      \Users\eminem\Desktop\galf.jpeg
0×3e8d19e0      \Users\eminem\Desktop\Screenshot1.png
0×3e8da630      \Users\eminem\Desktop\DumpIt
0×3e912190      \Users\eminem\Desktop\DumpIt\DumpIt.exe
0×3e9189d0      \Users\eminem\Desktop\DumpIt\DumpIt.exe
0×3e922070      \Users\eminem\Desktop\DumpIt
0×3eb5ad10      \Users\eminem\Desktop
0×3eba3e60      \Users\eminem\Desktop
0×3ebe2a20      \Users\eminem\Desktop\DumpIt\2PAC-20190629-072925.raw
0×3f9ffcb0      \Users\eminem\Desktop\DumpIt
0×3fc398d0      \Users\SlimShady\Desktop\Important.txt
0×3fcfdb00      \Users\SlimShady\Desktop
0×3fcfeac0      \Users\SlimShady\Desktop
```

```
┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f /home/kali/Downloads/INC-2026-DFIR-8941.raw -o ~/Downloads/inc_out windows.dumpfil
es --physaddr 0×3e8ad250
Volatility 3 Framework 2.28.3
Progress:  100.00               PDB scanning finished
Cache   FileObject      FileName        Result

DataSectionObject       0×3e8ad250      galf.jpeg       file.0×3e8ad250.0×fa80022ad0f0.DataSectionObject
.galf.jpeg.dat

┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ strings ~/Downloads/inc_filescan.txt | grep -i important
0×3f939720      \Users\SlimShady\AppData\Roaming\Microsoft\Windows\Recent\Important.lnk
0×3fc398d0      \Users\SlimShady\Desktop\Important.txt
```

```
┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ cd ~/Downloads/inc_out

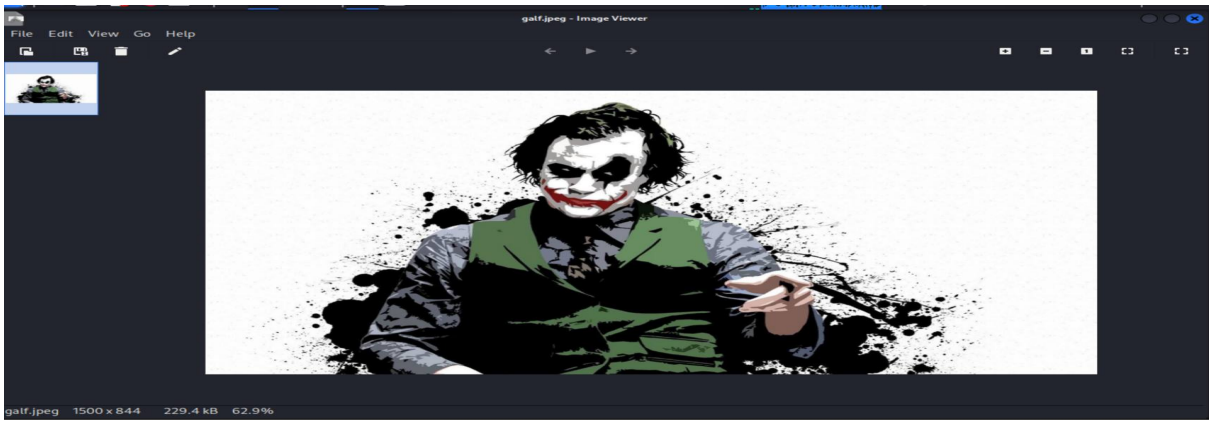
┌──(vol-env)─(kali㉿kali)-[~/Downloads/inc_out]
└─$ ls
file.0×3e8ad250.0×fa80022ad0f0.DataSectionObject.galf.jpeg.dat

┌──(vol-env)─(kali㉿kali)-[~/Downloads/inc_out]
└─$ cp *galf.jpeg*.dat galf.jpeg

┌──(vol-env)─(kali㉿kali)-[~/Downloads/inc_out]
└─$ file galf.jpeg
galf.jpeg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1×1, segment length 16, baseline,
precision 8, 1500×844, components 3

┌──(vol-env)─(kali㉿kali)-[~/Downloads/inc_out]
└─$ strings galf.jpeg | grep -iE "flag|inctf|ctf\{|key|pass"

┌──(vol-env)─(kali㉿kali)-[~/Downloads/inc_out]
└─$ xdg-open galf.jpeg
```




- بعدها استخدمت أمر filescan وبحثت عن مجلدات المستخدمين (eminem و SlimShady)، وطلعت لي ملفات سطح المكتب ومنها صورة galf.jpeg وملف Important.txt المحذوف. قدرت استخرج صورة galf.jpeg من الذاكرة، وفتحتها واستعرضت محتواها وطلعت صورة الجوكر.

```
┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f /home/kali/Downloads/INC-2026-DFIR-8941.raw windows.dumpfiles --pid 2432 2>/dev/nu
ll
Volatility 3 Framework 2.28.3

Cache   FileObject      FileName        Result

DataSectionObject       0×fa8000f40910  StickyNotes.snt file.0×fa8000f40910.0×fa8000f40590.DataSectionOb
ject.StickyNotes.snt.dat
ImageSectionObject      0×fa80015c3790  user32.dll      file.0×fa80015c3790.0×fa800151b3f0.ImageSectionO
bject.user32.dll.img
ImageSectionObject      0×fa80013d02b0  ntdll.dll       file.0×fa80013d02b0.0×fa80013c7950.ImageSectionO
bject.ntdll.dll.img
ImageSectionObject      0×fa80013c9810  kernel32.dll    file.0×fa80013c9810.0×fa800135ddf0.ImageSectionO
bject.kernel32.dll.img
ImageSectionObject      0×fa800151b300  msvcrt.dll      file.0×fa800151b300.0×fa80015197f0.ImageSectionO
bject.msvcrt.dll.img
ImageSectionObject      0×fa8001d74470  comctl32.dll    file.0×fa8001d74470.0×fa8001d986a0.ImageSectionO
bject.comctl32.dll.img
ImageSectionObject      0×fa8001deb700  dui70.dll       file.0×fa8001deb700.0×fa8001de9370.ImageSectionO
bject.dui70.dll.img
ImageSectionObject      0×fa8000f3b6f0  msxml3.dll      file.0×fa8000f3b6f0.0×fa8000f3e3c0.ImageSectionO
bject.msxml3.dll.img
ImageSectionObject      0×fa8000f095b0  StikyNot.exe    file.0×fa8000f095b0.0×fa8000f0d260.ImageSectionO
bject.StikyNot.exe.img
ImageSectionObject      0×fa8000f3eca0  InkObj.dll      file.0×fa8000f3eca0.0×fa8001cb6a40.ImageSectionO
```

```
┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ strings file.0×fa8000f40910.0×fa8000f40590.DataSectionObject.StickyNotes.snt.dat | grep -iE "flag|pa
ss|key|st3g|inctf|:"
{\*\generator Msftedit 5.41.21.2510;}\viewkind4\uc1\pard\tx360\tx720\tx1080\tx1440\tx1800\tx2160\tx2520\
tx2880\tx3240\tx3600\tx3960\tx4320\tx4680\tx5040\tx5400\tx5760\tx6120\tx6480\tx6840\tx7200\tx7560\tx7920
\tx8280\tx8640\tx9000\tx9360\tx9720\tx10080\tx10440\tx10800\tx11160\tx11520\highlight0\f0\fs22 The clipb
oard plugin works well but it doesn't give the flag :P\lang9\f1\par

┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ ls -la file.*StickyNotes*
-rw------- 1 kali kali 4096 Oct 17 14:20 file.0×fa8000f40910.0×fa8000f40590.DataSectionObject.StickyNote
s.snt.dat
```

```
┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ grep -i "Important.txt" ~/Downloads/inc_filescan.txt
0×3fc398d0      \Users\SlimShady\Desktop\Important.txt

┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ cd ~/Downloads

┌──(vol-env)─(kali㉿kali)-[~/Downloads]
└─$ { strings -a INC-2026-DFIR-8941.raw; strings -a -el INC-2026-DFIR-8941.raw; } | grep -aoiE "(flag|in
ctf|d4rk|ctf)\{[^}]{1,100}\}" | sort -u

┌──(vol-env)─(kali㉿kali)-[~/Downloads]
└─$ cd ~/Downloads/inc_out
```

```
┌──(vol-env)─(kali㉿kali)-[~/Downloads/inc_out]
└─$ sudo gunzip -k /usr/share/wordlists/rockyou.txt.gz && stegseek galf.jpeg /usr/share/wordlists/rockyo
u.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: ""
[i] Original filename: "LMAO.txt".
[i] Extracting to "galf.jpeg.out".


┌──(vol-env)─(kali㉿kali)-[~/Downloads/inc_out]
└─$ cat ~/Downloads/inc_out/galf.jpeg.out
Move on bro, there is nothing here :)
```

- استخدمت أداة stegseek على صورة galf.jpeg مع قائمة كلمات المرور واستخرجت ملف نصي مخفي (LMAO.txt). ولما قريت محتواه طلعت رسالة طعم (: Move on bro, there is nothing here

```
┌──(vol-env)─(kali㉿kali)-[~/volatility3]
└─$ cd ~/Downloads
F=INC-2026-DFIR-8941.raw
LC_ALL=C grep -aboP "I\x00m\x00p\x00o\x00r\x00t\x00a\x00n\x00t\x00\.\x00t\x00x\x00t\x00" "$F" | cut -d:
-f1 | while read off; do
  echo "=== offset $off ==="
  dd if="$F" bs=1 skip=$(( off>900 ? off-900 : 0 )) count=1800 2>/dev/null | strings -a
done
=== offset 310400668 ===
Wind
Wisp
View
-132
2917817172
N{59099400-57FF-11CE-BD94-0020AF85B590} {000214E4-0000-0000-C000-000000000046} 0×FFFFg
{4A7DED0A-AD25-11D0-98A8-0800361B1103} {000214E9-0000-0000-C000-000000000046} 0×FFFFx1
{55B3A0BD-4D28-42FE-8CFB-FA3EDFF969B8} {000214E9-0000-0000-C000-000000000046} 0×FFFF
{1F2E5C40-9550-11CE-99D2-00AA006E086C} {000214E9-0000-0000-C000-000000000046} 0×FFFF*
Important.txt.lnk
Important.txt.lnk
4ersion
StreamMRU
MRUListEx
ViewView2
1SPS
```







- بحث عن اسم الملف بترميز UTF-16 داخل الدمب عشان ألقى سجل الـ MFT الخاص فيه (علامة FILE0)، ولأن حجم الملف صغير كان محتواه مضمّناً (resident) بداخل السجل بعدها باستخدام أداة dd و strings طلعت أجزاء العلم مقسمة ومفصوله بأسطر فارغة متكررة كأنه أسلوب إخفاء تعمده المهاجم وكانت تطلع معي رسالة ختامية Good work :P.

وأخيرا بأوامر التنظيف والتجميع مثل tr لحذف الأسطر الفارغة و grep قدرت اضبط الأجزاء ولصقها ببعض وطبعت العلم النهائي كامل وصحيح .

{inctf{1_is_n0t_EQu4l_7o_2_bUt_th1s_d0s3nt_m4ke_s3ns3